

A Multifractal and Regime-Based Forecasting Framework for Brent Crude Oil Returns Using BDTI and Machine Learning

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Abstract

Financial time series, especially those of crude oil, have characteristics such as volatility clustering, heavy-tailed behavior, non-linear relationships, and structural changes. In this paper, a prediction framework for the Brent crude oil log-return using a multifractal and regime-based approach is introduced based on the lag-based indicators, BDTI, and multifractal parameters estimated by the Multifractal Detrended Fluctuation Analysis (MF-DFA). The difference between the largest and smallest scaling exponents in the multifractal spectrum ($\Delta\alpha$) measured through the rolling window analysis of 250 trading days reflects increasing complexity in markets. The dataset includes information from March 2013 to early 2026, containing 2,988 samples split into six regimes by using multi-signal binary segmentation. The four ensemble models, including Random Forest, XGBoost, LightGBM, and CatBoost, are compared using metrics such as MSE, RMSE, and MAE. The results reveal that multifractal characteristics remarkably boost the predictive power of the model, especially under high-volatility situations, in which XGBoost attains the maximum reduction of 64% of MSE in the COVID-19 regime. CatBoost produces the lowest error among other models (MSE = 0.000334). ANOVA shows the significance of $\Delta\alpha$ values.

Keywords: multifractal analysis; MF-DFA; Brent crude oil; Baltic Dirty Tanker Index; regime detection; machine learning; XGBoost; CatBoost; financial forecasting

1. Introduction

Financial market time series, including those associated with commodity markets such as crude oil, exhibit highly complex dynamics characterized by volatility clustering, fat tails, nonlinearity, and structural breaks [15–17]. Conventional econometrics models such as ARIMA and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) require linearity and stationarity, which are highly improbable to be present in real financial datasets. Therefore, these models cannot capture the complexity of financial market dynamics, especially in crisis and regime-changing times.

The dynamics of Brent crude oil are influenced by numerous factors, ranging from geopolitical tensions to an imbalance between supply and demand and logistical issues. In this sense, Baltic Dirty Tanker Index (BDTI) can be taken into account as one of the external factors influencing the dynamics of the global crude oil market because it presents a rate of transportation of the oil via ships. In this way, although the connections among the discussed indicators are highly complex, implying the presence of nonlinear structures that cannot be captured by conventional models.

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In order to find such complicated patterns, multifractal analysis methods, in particular, Multifractal Detrended Fluctuation Analysis (MF-DFA), first proposed by Kantelhardt et al. [1], are widely used in order to describe non-stationary time series. MF-DFA allows estimating the generalized Hurst exponent $H(q)$, which shows the scaling properties of the process. If a generalized Hurst exponent is strictly monotonically decreasing, multifractality exists which arises because of the presence of temporal correlation within the dataset. According to the work of Kwapień et al. [3], the multifractality of financial market time series is caused precisely by the temporal correlation and not by the fat-tailed distribution of returns.

The width of the multifractal spectrum can be calculated through a formula

$$\Delta\alpha = \alpha_{\max} - \alpha_{\min} \quad (1)$$

which demonstrates the heterogeneity of the scaling exponents. Higher values of $\Delta\alpha$ mean higher heterogeneity and market complexity. Numerous papers have revealed the multifractal properties of the energy market. For example, Chen et al. [4] found multifractal cross-correlations between crude oil prices and tanker freight rates, and Li et al. [5] demonstrated how the complexity of shipping freight rates grew during the outbreak of the COVID-19 pandemic.

According to the standard economic theory, there exists a causal relationship between oil prices and tanker freight rates because of the mechanism of cost operations [6]. Nonetheless, many recent works have refuted this point of view. According to the paper by Siddiqui and Basu [7], oil prices lead tanker freight rates for medium-term and long-term observations, while Shi et al. [8] state that non-supply oil price shocks play a greater role than tanker freight rates. Therefore, indices such as BDTI are suitable for being used as predictors. Chen et al. [2] demonstrated that BDTI return can be used as a leading indicator of Brent oil price during times of high market complexity. In particular, within the machine learning predictive framework proposed by [2], taking into consideration the multifractal nature of the market, they were able to predict Brent oil price accurately with a remarkable improvement in terms of performance (R^2 : 55–182%, MSE: 60–92%).

Ensemble machine learning frameworks such as gradient-boosted decision trees have proven to be superior approaches for financial forecasting purposes [13]. XGBoost [10], LightGBM [11], and CatBoost [12] were developed for solving the problem of structured financial forecasting with great success. In particular, thanks to their novel approaches (regularized gradient boosting, histogram-based tree learning, and ordered boosting with ordered target statistics), they have achieved remarkable performance. Moreover, since structural breaks and regimes are well known for commodities markets [9], applying the binseg algorithm enables identifying change-points efficiently in order to evaluate the performance of the model within the specific regimes. Recent advancements in deep learning have further demonstrated the effectiveness of data-driven approaches in handling complex, nonlinear patterns in real-world datasets [18].

Although the field has experienced notable progress recently, there is still room for development. For instance, most of the papers focus on price-level predictions, which seem inappropriate, taking into account the random-walk property of financial prices. Besides, multifractal properties are regarded as stationary, while changes within these properties need to be accounted for. Lastly, there is not enough research conducted about the influence of regimes and multifractality on the forecasting performance.

The four contributions of this research paper are as follows: (i) the sample is extended till 2026, covering the latest geopolitical incidents, (ii) use of log-returns rather than price level for forecasting, (iii) implementation of rolling multifractal analysis using MF-DFA; and (iv) incorporation of a regime detection algorithm.

The primary hypothesis of this paper states that the multifractal spectrum width $\Delta\alpha$ can act as a conditional predictor of model accuracy. Particularly, it is expected to have a significant role during volatile and extreme market regimes, as high complexity increases the predictive power of the model, while in the case of quiescent market regimes, it will have a modest role because such regimes can be considered as random walks.

The rest of this paper is organized as follows. Section 2 provides a description of the data used in this study. Section 3 describes the methodology that includes multifractal feature extraction via MF-DFA, regime detection by multisignal binary segmentation, feature construction, and machine learning modeling. Section 4 provides the results of the study, covering multifractal properties, model evaluation, feature selection, and regime and volatility-based analyses. The conclusion of this study is presented in Section 5.

2. Data Description and Preprocessing

The Baltic Dirty Tanker Index (BDTI) is the most commonly known index used to estimate the rates on tankers delivering crude oil along major international routes. This index is calculated according to the latest tanker market situation and includes data on supply, demand, vessel availability, and freight volumes. As the index covers real deals made in the oil shipping business, it may become a good instrument in analysing global oil market operations, providing some additional insights into this sphere compared to financial market prices.

On the other hand, Brent crude oil is another commonly referred to index price in terms of global oil benchmark and is applied for further economic analysis. Brent price takes into account market expectations and is affected by various geopolitical events, macroeconomic indicators, and the world oil market dynamics. Analysis of both indices allows for the investigation of financial and physical oil markets simultaneously.

For our research, we use a time series of daily values for the Brent price and BDTI covering the period from 14 March 2013 until 2 April 2026, which results in 3250 daily observations. Specifically, data on the BDTI were taken from Investing.com (composite index) from 2013 to 2023 and from the official Baltic Exchange website from 2023 to 2026. Time series of daily values for the Brent crude oil prices were retrieved from FRED (series: DCOILBRENTU). Data available from this source covers a variety of economic cycles and events, including an oil price crisis (2014-2016), the COVID-19 pandemic (2020), and recent geopolitical tensions between Russia and Ukraine since 2022. Thus, these events cause significant volatility and regime changes, providing an opportunity to study the multifractal dynamics of regimes. Figure 1 presents daily values of the Brent crude oil price and Baltic Dirty Tanker Index (BDTI) in the period of 2013-2026. One may note that there was high volatility in these time series caused by various market disruptions, including the global pandemic in 2020 and the Russia-Ukraine war that started in 2022.

2.1. Log-Return Calculation

In order to enhance the statistical behavior of the process and obtain stationarity, the prices are transformed into logarithmic returns:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \quad (2)$$

with P_t being the closing price at the time instant t . Log returns are better than simple returns because they add up over time and tend to have more symmetric distributions, which makes them good for multifractal analysis.

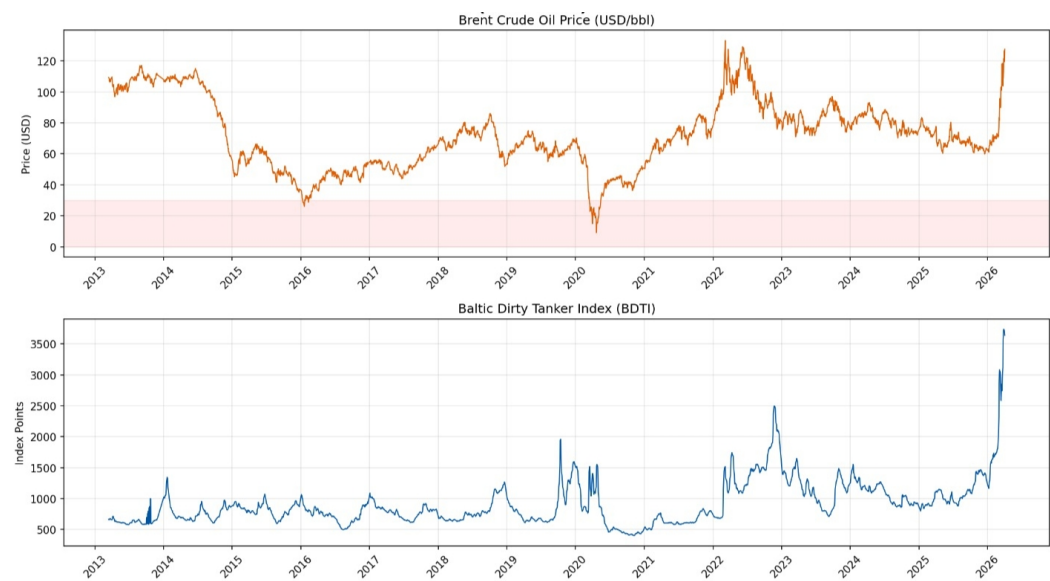


Figure 1. Daily time series of Brent crude oil and BDTI

2.2. Descriptive Statistics

127

Table 1. Descriptive statistics of the processed dataset (2013–2026)

Statistic	BDTI Return	Brent Return	$\Delta\alpha$
Count	2988	2988	2988
Mean	6.4×10^{-5}	-1.2×10^{-5}	0.691
Std. Dev.	0.024	0.028	0.279
Min	-0.38	-0.28	-0.340
Max	0.24	0.20	1.401

3. Methodology

128

The overall framework is organized in four stages: (i) Multifractal Feature Extraction Using Rolling MF-DFA; (ii) Market Regime Identification Using Multi-Signal Binary Segmentation; (iii) Feature Engineering with Three Increasingly Rich Configurations; and (iv) Performance Evaluation of Ensemble Models Under Regime-Based Stratification Relative to Baseline Methods. All computations are done through rolling windows to preclude lookahead bias.

3.1. Multifractal Detrended Fluctuation Analysis (MF-DFA)

135

The Multifractal Detrended Fluctuation Analysis (MF-DFA) method of Kantelhardt et al. [1] was employed for quantifying the scaling behavior of the returns time series. There are five steps in the MF-DFA process.

Step 1: Creation of the Profile Given a time series $x(i)$ of size N , its profile $Y(i)$ is defined as:

$$Y(i) = \sum_{k=1}^i [x(k) - \bar{x}], \quad i = 1, 2, \dots, N, \tag{3}$$

where $\bar{x}$ refers to the average of the time series.

142

Step 2: Profile Segmentation and Detrending The profile is partitioned into $2N_s$ non-overlapping segments, each of size s . For each segment v , a polynomial trend $P_v(i)$ (linear, $m = 1$ in this work) is subtracted:

$$F^2(v, s) = \frac{1}{s} \sum_{i=1}^s [Y((v-1)s + i) - P_v(i)]^2. \quad (4)$$

Step 3: Calculation of the q -Order Fluctuation Function For $q \neq 0$:

$$F_q(s) = \left\{ \frac{1}{2N_s} \sum_{v=1}^{2N_s} [F^2(v, s)]^{q/2} \right\}^{1/q}. \quad (5)$$

For $q = 0$:

$$F_0(s) = \exp \left\{ \frac{1}{4N_s} \sum_{v=1}^{2N_s} \ln [F^2(v, s)] \right\}. \quad (6)$$

q governs the relative contribution of the fluctuation sizes: large ($q > 0$) and small ($q < 0$). For this work, $q \in \{-5, -3, -1, 1, 3, 5\}$.

Step 4: Computation of the Generalized Hurst Exponent If scaling is present, $F_q(s)$ will exhibit the following relationship:

$$F_q(s) \sim s^{H(q)}, \quad (7)$$

where $H(q)$ is obtained by fitting a straight line on the logarithmic plot of $F_q(s)$ against s .

Step 5: Estimation of the Width of the Multifractal Spectrum With the Legendre transform, $\tau(q) = qH(q) - 1$, $\alpha = d\tau(q)/dq$, and $f(\alpha) = q\alpha - \tau(q)$, we can obtain the width of the multifractal spectrum as:

$$\Delta\alpha = \alpha_{\max} - \alpha_{\min}. \quad (8)$$

A large $\Delta\alpha$ reflects higher heterogeneity and complexity.

Step 6: Rolling Window Calculation of $\Delta\alpha$ In the rolling window implementation of MF-DFA, the length of the moving window is set to $W = 250$ observations, which slides over the time series by 5 observations at each step.

3.2. Regime Detection using Multi-Signal BinSeg Algorithm

In the course of structural shifts in the oil markets, the regime detection is implemented by the means of BinSeg algorithm that partitions the signal into segments according to multiple time-varying signals, specifically, Brent log-returns, rolling volatility, and $\Delta\alpha$ from the multifractal spectrum.

Binary segmentation implies a sequential approach in identifying change-points by splitting observations into subsequences where every time the most important breakpoint is found. Then the process is performed recursively in further split subsequences found in previous iterations up until there are no statistically significant breakpoints left.

For time series y_t , the identification of structural breaks is achieved through detection of change points τ minimizing segmentation cost:

$$\min_{\tau} [C(y_{1:\tau}) + C(y_{\tau+1:T})]$$

where $C(\cdot)$ represents a cost function. In this study, the following ℓ_2 cost function was utilized:

$$C(y_{a:b}) = \sum_{t=a}^b (y_t - \bar{y}_{a:b})^2$$

179

The use of multi-signal is due to the possibility of a change in other properties apart from returns, for instance, volatility. By considering few more variables we get a better estimation of the regime changes

The signals utilized for segmentation include:

- **Log Returns:** defined in Equation 2.
- **Rolling Volatility:**

$$\sigma_t = \sqrt{\frac{1}{w} \sum_{i=t-w+1}^t (r_i - \bar{r})^2}$$

186

where $w = 30$.

- **Multifractal Spectrum Width:** defined in Equation ??.

Binary segmentation is independently carried out for each signal resulting in a set of candidate breakpoints $\tau_k^{(r)}$, $\tau_k^{(\sigma)}$, and $\tau_k^{(\Delta\alpha)}$.

For our research, after conducting analysis, we have revealed six distinct regimes linked to various economical events, i.e., oil price crash period (2014-2016), pandemic

Table 2. Market regimes

R#	Period	Description	Vol	$\Delta\alpha$
1	2013–2014	Stability	0.011	0.688
2	2014–2018	Crash	0.023	0.753
3	2018–2020	Instability	0.022	0.656
4	2020–2021	COVID	0.059	0.592
5	2021–2022	Recovery	0.029	0.379
6	2022–2026	Uncertainty	0.022	0.769

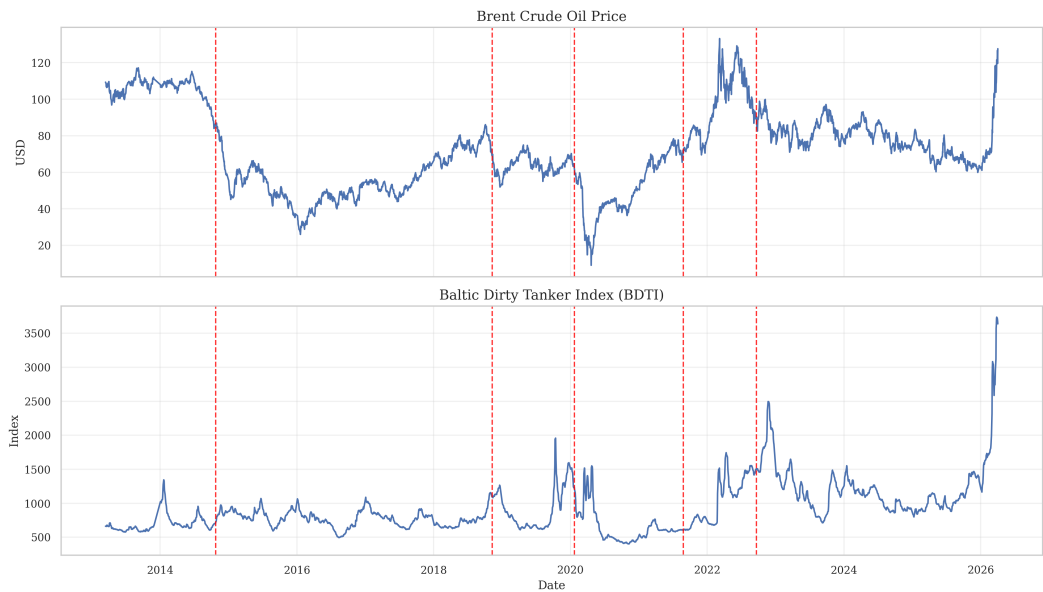


Figure 2. Market regimes identified using segmentation

3.3. Feature Engineering and Data Preparation193

The prediction target is the next-day Brent log-return:194

$$y_t = r_t^{\text{Brent}}.$$
(9)195

Three richer sets of feature configurations are created to allow us to detect the additional196
value added by each feature group197

Table 3. Feature configurations

Config	Features	Purpose
A	Brent Lag ₁	Baseline
B	Brent + BDTI	Cross-market
C	Brent + BDTI + Δα	Full model

All features are lagged by one period to avoid look-ahead bias. The dataset is split198
chronologically into training (80%) and testing (20%) sets.199

3.4. Machine Learning Models200

Four ensemble models are implemented:201

Random Forest: 300 trees with bootstrap sampling and feature subsampling.202

XGBoost [10]: minimizes203

$$L(\phi) = \sum_i \ell(\hat{y}_i, y_i) + \sum_k \Omega(f_k),$$
(10)204

where205

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|\omega\|^2.$$
(11)206

LightGBM [11]: uses Gradient-based One-Side Sampling (GOSS) and Exclusive Fea-207
ture Bundling (EFB).208

CatBoost [12]: employs ordered boosting to reduce prediction bias.209

Two baseline models are also considered: Linear regression and ARIMA/GARCH.210

3.5. Evaluation Metrics211

The performance of the models is measured using MSE, RMSE, and MAE. The coeffi-212
cient of determination (R^2) is not included since it does not offer much interpretation when213
dealing with near-random-walk data.214

The contribution of the change in α towards improving the model is calculated as215
follows:216

$$\text{Improvement (\%)} = \frac{\text{MSE}_B - \text{MSE}_C}{\text{MSE}_B} \times 100.$$
(12)217

A positive percentage means that there was improvement in the model, whereas a218
negative value indicates that additional noise was introduced. The figure 3.1 below shows219
the framework of the forecast methodology that we propose in this work. We have shown220
the sequence of steps (column on the left side in green), methods used in the respective221
step (column in blue in the middle), and the output from the process (column on the right222
side in gray). First, data are collected using BCTI and BDTI freight indices timeseries. In223
the data pre-processing phase, data normalization, data transformation, calculating log224
return values, rolling MF-DFA calculations to obtain multifractal spectrum width ($\Delta\alpha$).225
Multifractal analysis employs MF-DFA and MF-DMA techniques to extract long-memory226
characteristics and fractal nature. In feature engineering, many combinations of lagged227
return values, BDTI and $\Delta\alpha$ values, and break points information are formed. Regime228

detection is done using multi-signal binary segmentation on return values, volatility, and multifractal parameters. In the modeling phase, various machine learning ensembles such as random forest, XGBoost, LightGBM, and CatBoost are used with hyperparameter optimization. Finally, the model performance is evaluated using MSE, RMSE, and MAE per regime, and the model with optimal performance is selected.

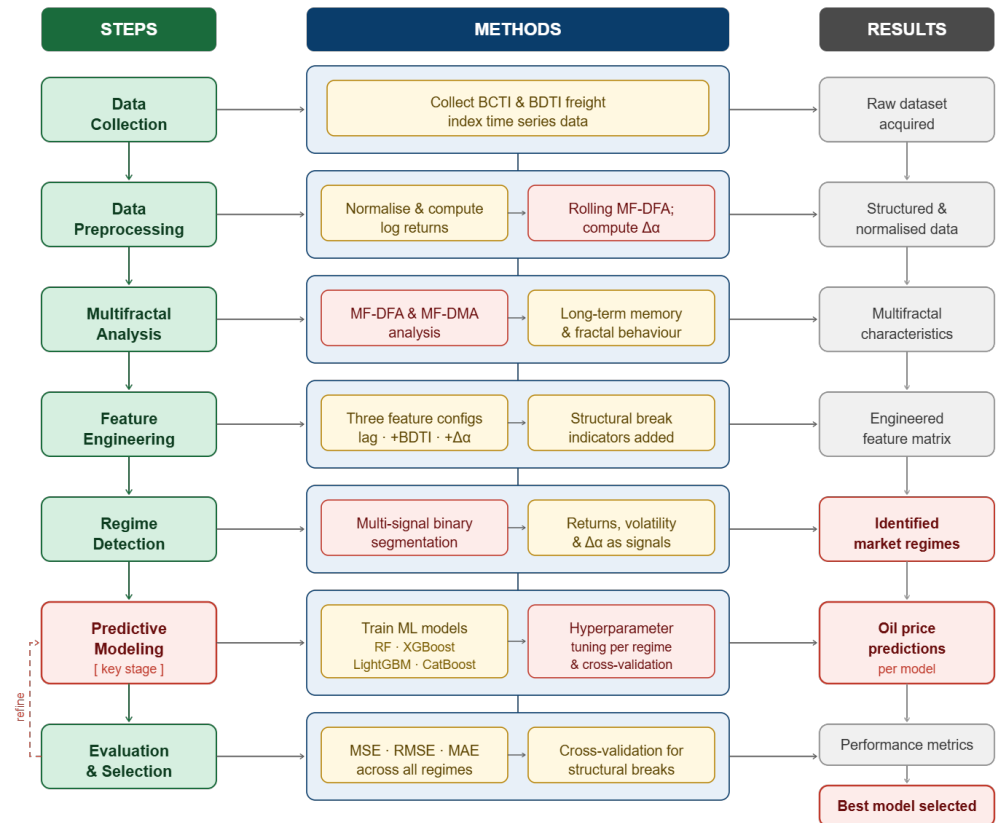


Figure 3. Forecasting Framework Methodology

4. Results and Discussion

4.1. Multifractal Characteristics of Brent and BDTI Returns

Multifractal characteristics of the series Brent and BDTI are studied by using the MF-DFA method. Fluctuation functions $F_q(s)$ for several orders $q \in \{-5, -3, -1, 1, 3, 5\}$ are shown in Figure 4. In case of both series, the fluctuation function manifests evident power law scaling properties, which is consistent with the existence of strong long-term dependencies.

Figure 4 shows fan-shaped divergence of the fluctuation function dependence with respect to q . The difference in the scaling exponent as a result of changing q indicates the multifractality of the series. Namely, positive values of q tend to amplify large fluctuations, while negative q values stress small fluctuations. Increasing divergence of the curves demonstrates that scaling behaviour in both series depends on q . MF-DFA fluctuation functions $F_q(s)$ for BDTI returns (left), and Brent returns (right) across $q \in \{-5, -3, -1, 1, 3, 5\}$. The fan-shaped divergence of slopes across q values confirms strong multifractal scaling behaviour in both series. In addition, Figure 5 displays generalized Hurst exponents $H(q)$ as a function of q . Since $H(q)$ decreases strictly with growing q , multifractality is demonstrated in both series, which implies that their scaling behaviour is heterogeneous with respect to different values of q . Generalized Hurst exponents $H(q)$ for BDTI (blue) and Brent crude

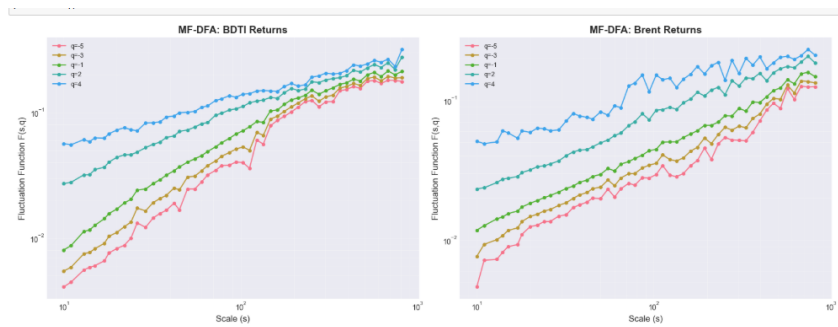


Figure 4. MF-DFA fluctuation functions $F_q(s)$ for BDTI returns and Brent returns

oil (orange) as a function of q . The strictly decreasing profiles confirm multifractality. The dashed line at ($H = 0.5$) represents the random walk benchmark.

The width of the multifractal spectrum is expressed as $\Delta\alpha = H(q_{\min}) - H(q_{\max})$. For the BDTI series, $\Delta\alpha$ equals 0.595, while for Brent returns $\Delta\alpha$ is 0.319. The value of the spectrum width of the considered cases greatly exceeds that in the case of a random walk model ($H = 0.5$).

Furthermore, since the multifractal spectrum width of BDTI is greater than that of Brent crude oil series, one can conclude that there is more diversity in the shipping freight market, which is reflected in high sensitivity to supply–demand mismatch and various structural shocks. Overall, multifractality is confirmed, and $\Delta\alpha$ is proved to be a relevant characteristic for the forecasting framework.

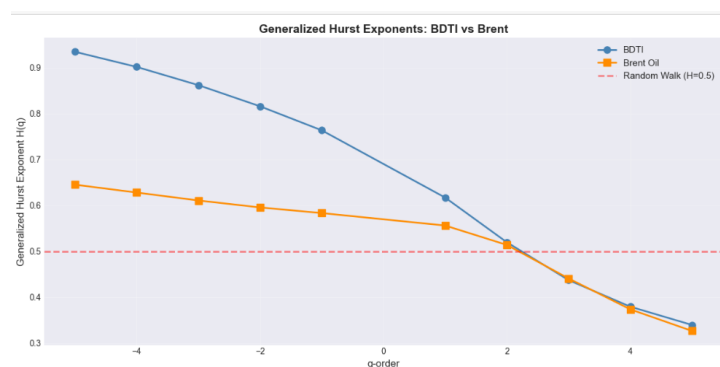


Figure 5. Generalized Hurst exponents $H(q)$ for BDTI and Brent crude oil as a function of q .

The width of the multifractal spectrum is given by $\Delta\alpha = H(q_{\min}) - H(q_{\max})$. It can be calculated to be 0.595 for the BDTI series and 0.319 for Brent returns. As seen, the values obtained significantly exceed the baseline for a random walk model ($H = 0.5$). It clearly demonstrates the existence of multifractal properties of both financial markets.

Additionally, a higher width of the multifractal spectrum indicates a more complex structure of the shipping freight market in comparison with the Brent crude oil market. It proves greater instability of the market as it depends on supply/demand imbalance and other factors. Figure 5 shows Generalized Hurst Exponents. The strictly decreasing profiles confirm multifractality. The dashed line at ($H = 0.5$) represents the random walk benchmark.

4.2. Model Performance

The overall test-set performance for all four machine learning models under full feature configuration (Configuration C) and linear regression, ARIMA, and GARCH baselines is presented in Table 4. It shows that the CatBoost model demonstrates the best overall performance with MSE equal to 0.000334, RMSE – 0.01828, and MAE – 0.01469. The reason

for it may be the ordered boosting algorithm used by CatBoost, as it eliminates the problem of prediction shifts in non-stationary financial time series.

All ensemble models outperform the ARIMA and GARCH baseline models, which confirms the superiority of nonlinear machine learning models for forecasting returns.

Table 4. Model comparison under full feature configuration (Configuration C)

Model	MSE	RMSE	MAE
CatBoost	0.000334	0.01828	0.01469
Random Forest	0.000364	0.01908	0.01533
LightGBM	0.000431	0.02077	0.01635
XGBoost	0.000478	0.02187	0.01737
Linear Regression	0.000505	0.02247	0.01593
ARIMA	0.000551	0.02347	0.01712
GARCH	0.000551	0.02347	0.01712

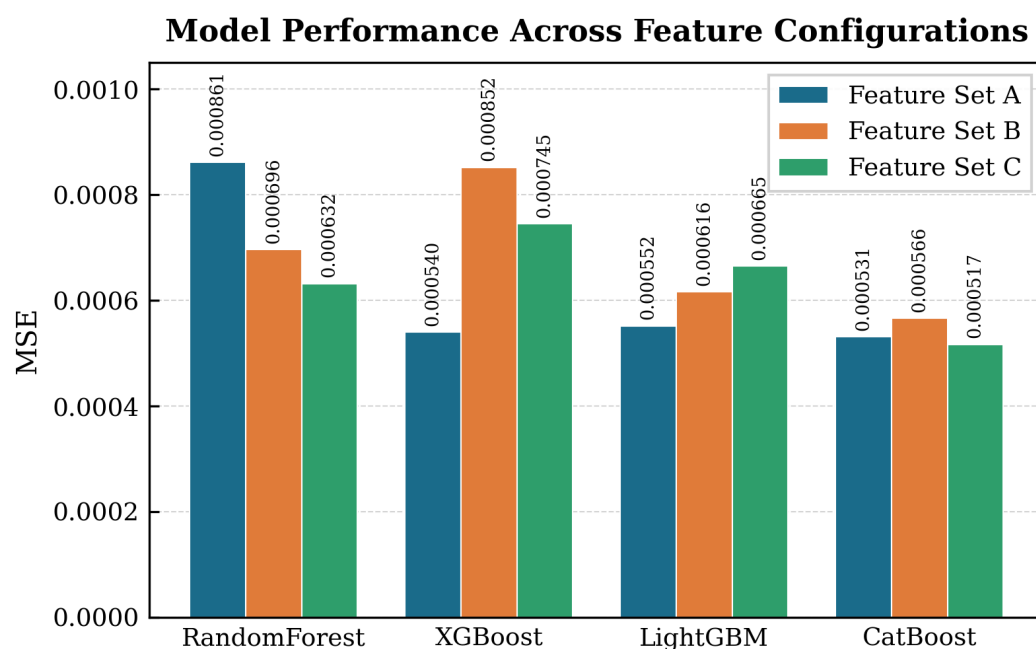


Figure 6. Model comparison across configurations

4.3. Feature Importance Analysis

The feature importance for Random Forest and XGBoost models is presented in Table 5. We can observe that the biggest proportion of the information comes from the lagged return of Brent (51.6%). In turn, 27.4% of information is provided by BDTI, and $\Delta\alpha$ accounts for 21.0%. In the case of XGBoost, a greater part of information is obtained from BDTI (40.3%) and $\Delta\alpha$ (36.4%).

Table 5. Feature importance comparison (Configuration C)

Feature	RF Importance	XGBoost Importance	Interpretation
Brent Lag ₁	0.516	0.232	Short-term momentum
BDTI Lag ₁	0.274	0.403	Cross-market signal
$\Delta\alpha$ Lag ₁	0.210	0.364	Market complexity

Feature Importance Comparison: Random Forest vs XGBoost

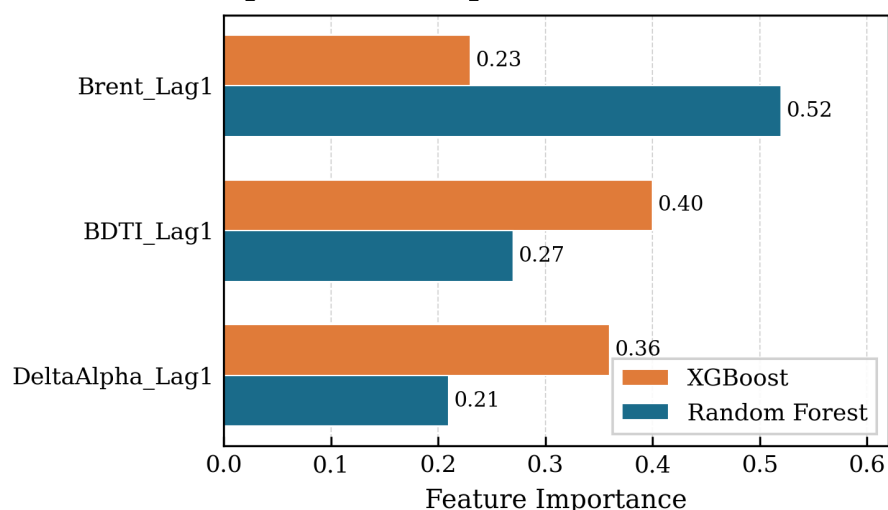


Figure 7. Model comparison across configurations

4.4. Regime-Wise Performance Analysis

One of the main contributions of this work lies in the analysis of the influence of adding the $\Delta\alpha$ variable in a regime-wise fashion. The percentage increase in MSE-based performance from Configuration B to Configuration C for each regime in the market is provided in Table 6

Table 6. Regime-wise MSE improvement (%) when adding $\Delta\alpha$

Regime	RF (%)	XGB (%)	LGB (%)	CAT (%)	Best
R2	+2.27	-13.21	-19.76	-12.21	RF
R3	+1.22	-14.66	-28.75	-1.50	RF
R4	+30.50	+64.09	-8.02	+12.54	XGB
R5	-5.35	-17.11	-17.55	+6.30	CAT
R6	+6.97	+3.09	+9.62	+7.06	LGB

The most impressive result was achieved in Regime 4 (COVID-19 crisis), where the XGBoost model had 64.09% of MSE reduced thanks to adding $\Delta\alpha$. Furthermore, it is linked to the maximum value of the volatility in Regime 4 (0.059) and a greater $\Delta\alpha$, implying that it has considerable discriminative power in the event of a high degree of multifractality in the market. In contrast, Regime 2 and Regime 3 demonstrate almost no positive effects, because during stable times, the presence of multifractal features in the market is not very important, and hence, their role is marginal. Regime 5 also demonstrates negative improvements due to transitional processes that generate noise.

4.5. Volatility-Stratified Analysis

To support the results obtained from the analysis of the regimes, the observations are divided in two groups – high and low volatility periods defined by the median rolling volatility level. In this section, the performance of the Random Forest model is considered. It is shown in Table 7.

From the results we can see that $\Delta\alpha$ enhances prediction accuracy irrespective of volatility, particularly at high levels of volatility

Table 7. Performance across volatility regimes (Random Forest)

Volatility	Config.	MSE	RMSE	MAE	MSE Impr. (%)
Low	B	0.000291	0.01706	0.01340	–
Low	C	0.000274	0.01656	0.01298	+5.84
High	B	0.001157	0.03401	0.02499	–
High	C	0.001085	0.03294	0.02388	+6.22

4.6. Discussion

From the obtained findings, we can conclude that multifractality characteristics work as conditional amplifiers which have their usefulness depending on the regime they operate under.

First, during crises, such as the current COVID-19 crisis, multifractal complexity, in the form of $\Delta\alpha$ grows because of market dynamics becoming more diverse.

Second, at times of stability, market dynamics are similar to random walks. Thus, the signal/noise ratio in multifractals decreases. This leads to insignificant or even negative effects of $\Delta\alpha$.

Third, the differences between algorithms show the ability of XGBoost to take advantage of the nonlinear relationship between multifractality measures, while the sampling method used by LightGBM makes the model prone to tail effects.

Consequently, the study results demonstrate the usefulness of using multifractals in market transitions.

5. Conclusions

This study proposes a multifractal and regime-based forecasting framework for predicting Brent crude oil log-returns. The dataset used in the research includes 2,988 daily observations from March 2013 to 2026. Six regimes were detected via multi-signal binary segmentation by means of returns, rolling volatility, and rolling $\Delta\alpha$ series. Below are the major conclusions drawn from the findings of the current paper.

First, multifractals represented by the spectrum width $\Delta\alpha$ increase forecasting efficiency when applied in stressful situations, while in regimes characterized by market stability their contribution becomes insignificant or even negative. The main contribution is reflected by a significant decrease in the mean squared error (MSE) – up to 64.09% for XGBoost during the period of the COVID-19 crisis.

Second, compared to the benchmark models such as ARIMA and GARCH, CatBoost demonstrates the highest performance in terms of MSE (0.000334), root mean squared error (RMSE) (0.01828), and mean absolute error (MAE) (0.01469).

Thirdly, the importance of features demonstrates that multifractal properties contribute to model accuracy by 21%-36%. This indicates that multifractality offers additional insights aside from autocorrelation in returns and signals between the markets generated from the shipping index. Also, the results of the ANOVA test prove the difference of the $\Delta\alpha$ values across regimes ($F = 121.11$, $p \approx 5.86 \times 10^{-117}$).

Fourthly, results from cross-correlation analysis indicate that there is near-zero linear correlation between the returns on Brent crude oil and BDTI across most regimes. This means that the dependency between the two variables is mostly nonlinear.

In conclusion, it can be said that the proposed methodology proves the hypothesis of the regime-dependency of financial predictability. Besides, multifractals proved to be valuable during stressful regimes as conditional amplifiers of predictive power.

5.1. Future Research Directions

There are several directions for future research related to the presented work. First, the application of deep learning techniques, such as LSTM or Transformer models with attention mechanisms can be extended by considering rolling $\Delta\alpha$ as a time-varying feature.

Second, online regime detection by means of the change-point detection methods including Bayesian approaches would allow conducting real-time adaptive forecasts.

Third, the feature space may be extended to include additional shipping indices, macroeconomic indicators, and alternative data sources such as satellite-based vessel tracking.

And, finally, the combination of various predictive models by means of ensemble stacking techniques would allow taking advantage of the individual strengths of each of them.

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